Bambu Lab P2S 3D Printer Safety Policy

1. Purpose

This policy defines the mandatory safety rules for operating the Bambu Lab P2S 3D printer in the lab. It is intended to prevent injury, fire, property damage, equipment damage, and hazardous exposure to fumes, particulates, hot surfaces, moving parts, sharp tools, or electrical hazards.

Every employee, contractor, intern, visitor, or other person who operates, assists with, cleans, maintains, or observes operation of the printer must follow this policy and sign the acknowledgement before participating in 3D printing work.

2. Scope

This policy applies to:

- 3D printing on the Bambu Lab P2S using approved filament materials.
- Setup, slicing, file transfer, filament loading, print starting, print monitoring, part removal, cleaning, troubleshooting, and routine maintenance.
- Use of the printer, build plate, hotend, nozzle, toolhead, filament path, AMS or filament handling accessories, printer software, and related hand tools.
- Anyone working near the printer while it is powered, heating, printing, cooling, being cleaned, or being serviced.

This policy does not cover laser cutting, laser engraving, resin printing, chemical processing, soldering, reflow, machining, or wet chemistry, which require separate procedures.

3. Conservative Safety Assumption

The person handling the Bambu Lab P2S shall treat it as a powered machine with hot components, moving parts, electrical circuits, and potential airborne emissions.

The printer shall not be treated as harmless because it is enclosed or automated. The enclosure, sensors, software, camera, filament detection, and automatic pause features are aids only. They do not replace operator training, monitoring, inspection, ventilation, or emergency response.

If the printer is damaged, behaving unexpectedly, producing smoke, producing a burning smell, making abnormal noise, showing repeated errors, or missing any required guard or cover, it must be removed from service until inspected and cleared.

4. Main Ways a Person Could Be Hurt

Employees must understand that injury can occur during routine printing, print removal, filament changes, cleaning, and maintenance. The principal hazards are:

1. **Burns from hot surfaces.** The nozzle, hotend, heat block, build plate, recently printed parts, molten filament, and purge material can burn skin.
2. **Pinch, crush, or entanglement injury.** The moving toolhead, belts, pulleys, motors, fans, lead screws, build plate, filament feed path, and AMS or spool mechanisms can pinch, strike, trap, or entangle fingers, hair, clothing, lanyards, or jewelry.
3. **Cuts, punctures, and scrapes.** Scrapers, blades, flush cutters, pliers, nozzle tools, broken prints, sharp supports, build-plate edges, and filament ends can cut or puncture skin.
4. **Eye injury.** Snapped filament, flying support material, broken plastic, tool slips, and debris from part removal or post-processing can injure eyes.
5. **Respiratory irritation or hazardous exposure.** Heated plastics can release fumes, odors, ultrafine particles, and volatile compounds. Overheated, unknown, contaminated, or unsuitable materials can produce more hazardous emissions.
6. **Fire or smoke.** Electrical faults, overheated components, clogged nozzles, failed prints, incorrect temperatures, combustible clutter, unsuitable materials, or unattended abnormal operation can cause smoke or fire.
7. **Electrical shock.** Damaged cords, wet hands, liquids near the printer, exposed wiring, improper repairs, modified electronics, or opening electrical compartments can create shock hazards.
8. **Chemical exposure.** Isopropyl alcohol, adhesives, solvents, lubricants, cleaners, filament additives, and contaminated debris can irritate skin, eyes, or lungs.
9. **Ergonomic strain.** Lifting the printer, moving filament boxes, leaning into the machine, repetitive scraping, or awkward part removal can cause strain.
10. **Unexpected operation.** Incorrect files, wrong material profiles, wrong temperatures, remote print starts, software errors, cloud or network commands, automatic calibration, or resume commands can cause unexpected heating or motion.
11. **Complacency risk.** Injuries are more likely when an operator assumes a familiar print is safe, leaves a failed print running, bypasses safety features, skips PPE, uses unknown filament, or ignores early warning signs.

5. Operator Responsibilities

5.1 Person Handling the Printer

The person handling the printer shall:

- Complete required training before use.
- Follow this policy, the manufacturer's manual, and the lab operating procedure.

- Confirm the printer, build plate, filament, print file, temperature settings, ventilation, and work area are safe before use.
- Wear required PPE and require nearby observers to follow safety instructions.
- Monitor the printer according to this policy.
- Stop work immediately if unsafe conditions exist.
- Report injuries, near misses, smoke events, fire events, equipment faults, abnormal odors, repeated errors, damaged parts, or unsafe conditions.
- Remove the printer from service if any required safety feature, cover, cable, fan, sensor, heating component, or motion component is damaged, missing, unreliable, or unclear.
- Avoid new materials, unusual temperature settings, hardware changes, firmware changes, electrical work, or non-routine maintenance unless the hazards are understood and the safe procedure is clear.

5.2 Observers and Visitors

Observers and visitors may approach the printer only with permission from the person handling the printer. They must follow all PPE, distance, and viewing requirements. Visitors may not operate the printer.

6. Authorization and Training

No person may operate the printer independently until all of the following are complete:

- Review of this policy and signature of the acknowledgement.
- Review of the current manufacturer manual and lab operating procedure.
- Hands-on training by a person already competent with this printer and procedure.
- Demonstrated ability to inspect the printer, verify ventilation, select approved filament, load filament safely, start and stop a print, monitor a print, pause or cancel a failed print, remove a print safely, and use emergency procedures.
- Demonstrated understanding of hot surfaces, moving parts, approved materials, prohibited materials, required PPE, fire response, and incident reporting.

Refresher review is required after any incident, near miss, equipment change, material change, process change, or extended period without use.

7. Approved Materials and Ventilation

Only approved filament materials may be used. The operator must confirm the filament type, temperature range, material profile, storage condition, and safety requirements before printing.

The printer must be used in a well-ventilated area. Fume extraction or filtration must be used when required by the material, lab procedure, or safety lead.

The following are prohibited unless specifically approved by the person responsible for the lab or equipment:

- Unknown, unlabeled, contaminated, wet, brittle, or damaged filament.
- Filament outside the printer's rated capability or outside the approved lab material list.
- Materials that produce strong fumes, smoke, charring, or unknown decomposition products.
- Materials requiring ventilation, filtration, chamber temperatures, nozzle temperatures, or bed temperatures that the lab setup cannot safely provide.
- Filament or spool setups that bind, tangle, scrape, jam, or interfere with the printer's motion.

If a material produces smoke, strong odor, visible residue, abnormal extrusion, or unexpected printer behavior, stop the print and report the issue.

8. Required Engineering Controls

The following controls must be in place before operation:

- The printer must be placed on a stable, level, nonflammable or fire-resistant surface.
- The printer must have adequate clearance for ventilation, cable routing, doors, spool movement, and maintenance access.
- The enclosure, doors, covers, panels, fans, and cable routing must be intact and properly installed.
- The power cord, plug, outlet, and any surge protection must be undamaged and appropriate for the printer.
- The printer area must be free of paper, cardboard, wipes, solvents, aerosols, loose plastic scraps, cloth, and unnecessary combustible material.
- A suitable fire extinguisher must be visible, accessible, inspected, and appropriate for the likely fire class in the lab.
- Required ventilation or filtration must be operating before printing and must continue as needed after printing to clear odors and emissions.
- Tools for print removal and maintenance must be stored safely and used only for their intended purpose.

If any required control is missing or defective, the printer must not be operated.

9. Required PPE

Minimum PPE for normal operation:

- Safety glasses when loading filament, cutting filament, removing prints, removing supports, cleaning the printer, or working near the toolhead.
- Heat-resistant gloves when handling the build plate, hot parts, hot purge material, the nozzle area, or recently printed parts.

- Cut-resistant gloves when using scrapers, blades, flush cutters, deburring tools, or when removing sharp supports.
- Nitrile gloves when handling adhesives, solvents, isopropyl alcohol, lubricants, cleaning chemicals, or contaminated debris.

Operators must secure long hair, loose clothing, jewelry, lanyards, hoodie strings, and dangling items before working near the printer.

PPE does not make it safe to touch the nozzle, hotend, moving toolhead, belts, fans, or energized electrical parts.

10. Pre-Use Checklist

Before each print, the operator shall:

1. Confirm they are trained and not impaired, distracted, or rushed.
2. Confirm the printer is clean, undamaged, and free of loose tools, loose plastic, and combustible clutter.
3. Inspect the power cord, plug, enclosure, doors, fans, build plate, hotend area, nozzle, filament path, spool holder, and AMS or filament accessory if used.
4. Confirm the build plate is installed correctly, clean, and suitable for the selected material.
5. Confirm the filament is approved, dry enough for use, correctly identified, and loaded without tangles or binding.
6. Confirm the print file, slicer settings, material profile, nozzle temperature, bed temperature, support settings, and estimated print time are appropriate.
7. Confirm required ventilation or filtration is operating.
8. Confirm the fire extinguisher is accessible.
9. Confirm no one has hands, tools, hair, clothing, or personal items inside the printer.
10. Confirm the first layer can be monitored before leaving the printer area.

If anything is uncertain, stop and do not operate the printer until the safe setup and procedure are clear.

11. Operating Rules

Operators shall follow these rules during every print:

- Monitor the start of every print, including bed leveling, calibration, purging, and first-layer adhesion.
- Do not reach into the printer while the toolhead, bed, fan, filament feed, or AMS is moving.
- Do not touch the nozzle, hotend, heat block, build plate, molten plastic, or freshly printed part unless it has cooled or proper heat protection is used.

- Do not open panels, remove covers, bypass sensors, defeat interlocks, or modify safety features.
- Do not use phones, headphones, unrelated computer work, or other distractions in a way that prevents active monitoring during setup and first-layer verification.
- Do not leave a print running if there is poor bed adhesion, spaghetti failure, a clogged nozzle, scraping, smoke, burning smell, abnormal noise, severe vibration, or repeated error messages.
- Do not spray cleaners, aerosols, or solvents near hot components.
- Do not place drinks, liquids, wet wipes, or chemical containers on or near the printer.
- Do not use the printer as a storage shelf.
- Do not force a stuck print off the build plate in a way that directs a scraper or blade toward the body.
- Do not leave the printer powered, heated, or remotely accessible to untrained users when not in active use.

Stop the print immediately if it behaves differently than expected.

12. Remote Operation

Remote monitoring may be used as an additional control, but it is not a substitute for safe setup and physical inspection.

Remote print starts are allowed only when:

- The printer area has been inspected before the print.
- The correct material and build plate are loaded.
- The operator can monitor the print and respond to faults in a reasonable time.
- The print does not involve an untested material, unusual settings, or elevated risk.

Remote starts are prohibited if the printer has not been inspected, if combustible clutter is present, if ventilation status is unknown, if the material is unfamiliar, or if no trained person can respond to an abnormal condition.

13. Part Removal and Post-Processing

After a print, the operator shall:

1. Confirm the printer has stopped moving.
2. Allow the build plate and part to cool unless heat-resistant gloves are used.
3. Remove the build plate carefully if required by the lab procedure.
4. Flex removable build plates only as intended by the manufacturer.
5. Use scrapers, blades, and cutters carefully, directing force away from the body.
6. Wear safety glasses when removing supports or cutting filament.

7. Dispose of purge waste, failed prints, support material, filament ends, and plastic debris according to lab procedures.
8. Clean the printer area and return tools to storage.

Printed parts may have sharp edges, brittle supports, hot internal areas, or residue from adhesives or filament additives.

14. Maintenance and Troubleshooting

Routine maintenance may be performed only by trained users. Non-routine maintenance, electrical work, firmware modification, disassembly beyond normal user maintenance, or repair must be performed only by a person competent for that task under a specific safe procedure.

Before maintenance, the operator shall:

- Stop the print.
- Allow hot components to cool unless heat is required by the approved procedure.
- Power down and unplug the printer when required by the maintenance task.
- Keep hands clear of moving parts.
- Use proper tools and PPE.

During troubleshooting:

- Never pull hardened plastic from the hotend with bare hands.
- Never clear a jam while the printer is moving.
- Never touch exposed wiring or internal electronics.
- Never operate the printer with guards, covers, fans, or safety features removed except under an approved service procedure.
- Tag the printer out of service if a fault cannot be corrected safely.

15. Housekeeping and Waste

The printer area must be kept clean and orderly.

Operators shall:

- Remove scrap plastic, purge waste, failed prints, support material, and filament clippings after use.
- Keep tools, blades, scrapers, and cutters stored safely.
- Keep adhesives, solvents, cleaners, lubricants, and isopropyl alcohol capped and labeled.
- Keep combustible materials away from hot components and electrical equipment.
- Dispose of contaminated wipes, chemical waste, and damaged parts according to lab waste procedures.

Do not blow plastic dust, fibers, or debris into the room with compressed air.

16. Emergency Procedures

16.1 Burns

- Move away from the hot source.
- Cool the burn with clean cool water according to first-aid procedure.
- Do not pull hardened plastic from skin if it is stuck.
- Report the injury immediately.
- Seek medical attention for serious burns, burns involving molten plastic, or burns larger than a minor superficial injury.

16.2 Cuts, Punctures, or Eye Injury

- Stop work.
- Use first aid as appropriate.
- Seek medical attention for eye injury, deep cuts, embedded debris, or uncontrolled bleeding.
- Report the injury immediately.

16.3 Pinch, Crush, or Entanglement

- Stop the printer immediately.
- Use the printer controls or power disconnect if needed and safe.
- Do not restart the printer until the person is clear and the cause is understood.
- Report the incident immediately.

16.4 Fire, Smoke, or Burning Smell

- Stop the print immediately.
- Disconnect power only if safe to do so.
- Alert nearby personnel.
- Use a fire extinguisher only if trained, the fire is small, and there is a clear exit path.
- Evacuate and call emergency services if the fire grows, produces heavy smoke, involves electrical equipment, or cannot be controlled immediately.
- Do not resume operation until the printer has been inspected and the cause has been corrected.

16.5 Fumes, Strong Odor, or Ventilation Failure

- Stop the print immediately.
- Keep the enclosure closed if safe to contain emissions.
- Leave the area if fumes escape into the room or cause irritation.
- Notify the person responsible for the lab or equipment.
- Do not resume until ventilation is restored and the cause is understood.

16.6 Electrical Fault

- Stop work.
- Do not touch exposed wiring, damaged cords, wet equipment, or energized parts.
- Disconnect power only if safe to do so.
- Tag the printer out of service.
- Notify the person responsible for the lab or equipment.

17. Stop-Work Authority

Every employee has authority to stop 3D printing work if they believe conditions are unsafe. Work may resume only when the unsafe condition is corrected and the person handling the printer has confirmed the setup is safe.

18. Employee Acknowledgement

By signing below, I acknowledge that:

- I have read and understood this Bambu Lab P2S 3D Printer Safety Policy.
- I understand the ways the printer can injure me or others, including burns, pinch injuries, cuts, eye injury, fumes, fire, electrical hazards, chemical exposure, and ergonomic strain.
- I agree to operate the printer only when trained, unimpaired, and following the safe procedure.
- I agree not to bypass safety controls, operate damaged equipment, use unsuitable materials, leave abnormal prints running, or allow untrained operation.
- I agree to report incidents, near misses, equipment faults, unsafe conditions, smoke, fire, abnormal odors, and injuries immediately.

Employee name: ______________________________

Employee signature: ___________________________

Date: _______________________________________

19. References for Safety Basis

- Manufacturer manual, labels, specifications, and maintenance instructions for the exact Bambu Lab P2S unit installed in the lab.
- Safety data sheets and supplier information for filament materials, adhesives, cleaners, lubricants, and solvents used in the lab.
- OSHA general industry safety principles for machine guarding, electrical safety, fire prevention, hazard communication, and personal protective equipment.
- NIOSH and other occupational health guidance regarding emissions from heated thermoplastics and additive manufacturing processes.

Signature

Printed Name	Signature
Date	Supervisor / Reviewer